APPENDIX "A"

PRIME CONTRACTOR'S NOTICE OF UNSAFE OR HARMFUL CONDITION OR ACT (the "Notice")

This Notice has been issued by the Prime Contractor to:

Name of Contractor: _____

Address: _____

The following unsafe or harmful condition or act was observed at the following location and time:

Description of unsafe or harmful condition or act: _____

_____

Location: _____ Date: _____ Time: _____ AM/PM

The Contractor named in this Notice is required to correct the above referenced unsafe or harmful condition or act prior to performing any more work activity at the location identified above.

Prime Contractor: _____

Signature: _____ Date: _____ Time: _____ AM/PM

Receipt acknowledged by the Contractor:

Signature: _____ Date: _____ Time:_____ AM/PM

The Ministry Manager received a copy of this Notice on: Date: _____ Time: _____ AM/PM

On the basis of this Notice, the Ministry Manager will issue an "Stop Work Order" in connection with the unsafe or harmful condition or act as identified on this Notice.

The Prime Contractor will, if satisfied that the unsafe or harmful condition or act has been rectified cancel and withdraw this Notice.

Date of cancellation/withdrawal: _____ Time: _____ AM/PM

Prime Contractor - Signature: _____

The Ministry Manager will issue an "Resume Work Order" when in receipt from the Prime Contractor of this Notice endorsed to indicate that the unsafe or harmful condition or act has be rectified.

DISTRIBUTION: Original: Contractor Copy: Ministry Manager File Copy: Prime Contractor

APPENDIX "B"

BRITISH COLUMBIA Ministry of Transportation

CONTRACTOR'S ACCIDENT STATISTICS SUMMARY REPORT

Project Number _____ Date (yyyy/mm/dd) _____

Description _____

_____

_____

Award Date (yyyy/mm/dd) _____ Completion Date (yyyy/mm/dd) _____

Name of Contractor _____

Month	Total Number of All Hours Worked by All Contractor's & Sub-contractor's Workers	Number of Reportable Accidents & Incidents	Number of Days Lost Due to Lost Time Accidents	Total Cost for Damage to Equipment or Material
Jan				
Feb				
Mar				
Apr				
May				
Jun				
Jul				
Aug				
Sep				
Oct				
Nov				
Dec				
Totals				

SAMPLE

Contractor's designated authority - Signature _____

Print name of Contractor's designated authority _____

Attach additional copies of this form as necessary and submit to Ministry Representative upon contract completion.

H1024 (1997/08)

Page _____ of _____

SECTION 145

GENERAL REQUIREMENTS FOR HIGHWAY, BRIDGE AND MARINE CONSTRUCTION

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

145.01 General Requirements - This Section covers the general Work requirements for highway, bridge, and marine construction.

The standards and materials specifications contained in the Design Build Standard Specifications govern the specific requirements for the Work. The Contract may provide additional specific requirements for the Work and supersede the Design Build Standard Specifications where there is conflict.

Specifications, standards, regulations and codes of other authorities named in full or by acknowledged abbreviations in the Glossary of Terms, the Design Build Standard Specifications or the Contract Document Package shall be adopted by reference.

Such reference in whole or in part by the necessary alphanumerical designation from the latest edition or revision is intended to fully detail the requirements of the Work.

145.02 Reference Information - Reference information for the Work identified in the Contract is made available for the Contractor to use for its own purpose. The Contractor shall be solely responsible for any interpretation placed thereon and for all impacts resulting from such interpretation.

145.03 Value Engineering Proposals - The Ministry wishes to encourage the development of value engineering proposals (VEP's) during the Term of the Contract. The Ministry will provide a process, as outlined in DBSS 125 to encourage the introduction, consideration, and possible implementation of VEP's. The Contractor may submit any number of VEP's during the Term.

The Ministry will be the sole judge of the acceptability of a VEP and of the cost savings, which will be realized by the Ministry due to the implementation of the VEP. The decision of the Ministry will be final.

The Ministry will not be liable to the Contractor for the Ministry's failure to accept or implement a VEP submitted by the Contractor nor for any delays to the Work that may be attributable to the VEP. A VEP or Summary VEP cannot, under any circumstances, be the basis for a claim by the Contractor.

145.04 Scope of Work - The scope of Work is described in the Contract. No warranty is given or implied that any or all of this Work will be required or supplied.

145.05 Not Used

145.06 Not Used

145.07 Pre-construction (Post-Award) Meeting - The Ministry Representative will determine the pre-construction (post-Award) meeting's location and time and notify the Contractor. The meeting shall be held reasonably soon after Award. And afford both parties an opportunity to establish relationships and to gain an enhanced understanding of the project, the design, and each parties expectations.

145.08 Mobilization - Mobilization consists of the necessary Work and operations including, but not limited to, the movement of personnel, equipment, supplies, and incidentals to the Site, the establishment of offices, camps, and other facilities necessary to undertake the Work and all other Work Items and operations which must be initiated and finished as part of completion of the Work.

145.09 Authority of the Ministry Representative - The Ministry Representative and Ministry Manager have the authority to accept or reject the Contractor's Work in accordance with the conditions set out in the Contract. Until the Work meets the requirements of the Contract and has been accepted by the Ministry Representative, the Ministry is not obligated to make payment for Work done by the Contractor.

145.10 Power, Telegraph, Telephone Lines and Other Utilities - Moving of existing power, telegraph and telephone lines, poles and other utilities from their current to their final design locations is part of the Work.

If the Contractor's operations require the temporary removal or relocation of lines or poles, such removal or relocation shall be entirely the responsibility of the Contractor and costs of any interim utility relocations shall be to the Contractor's account. Utility owner(s) must be contacted prior to doing any utility work.

The Contractor shall be responsible for any damage to poles, lines or utilities caused by the Contractor's

construction operations, and shall repair and make good the same or bear the expense of making good and repairing the same.

The Contractor shall take the necessary precautions to protect existing utility services during the performance of the Work and shall be responsible for any losses or damages caused by the Contractor.

Locations of the utilities shown on the Drawings are approximate and may not be complete. The Contractor shall be responsible to ensure the locations of the utilities are complete and shall notify the Designer and the Ministry Representative of any discrepancies between the Drawings and the Contractor's findings.

145.11 Permits - The Contractor shall give all requisite notices in connection with the Work to the proper authorities and shall procure at the Contractor's expense all permits, licences, etc., of every description necessary for the construction and completion of the Work.

The Contractor shall deliver or make available all original certificates to the Ministry Representative for all or any part of the Work for which such certificates may be required in connection with performing the Work.

145.12 Inspection of Work - All materials and each part or detail of the Work shall be subject to inspection by the Ministry Representative.

The Contractor shall be held strictly to the true intent of the Contract in regard to quality of materials, work practice and the diligent execution of the Contract.

The Ministry Representative shall be given access to all or any part of the Work, and to the preparation, fabrication or manufacture of the materials to be used. The Contractor shall furnish all information and assistance as is required to make a complete, true and detailed inspection.

The Ministry Representative or Ministry Manager may reject any part or parts of the Work which do not comply with the requirements of the Contract.

Inspection may extend to all or any part of the Work and to the preparation, fabrication or manufacture of the materials to be used. The Ministry may station others on the Site to report on the progress of the Work and the manner in which it is being performed.

Where detected through Ministry Quality Audits, the Ministry Representative will report to the Contractor on materials furnished or Work performed that fails to fulfill the requirements of the Contract.

Inspection or its lack shall not relieve the Contractor from any obligations to furnish acceptable materials or to provide completed construction Work that complies with the Contract.

The Contractor shall furnish for the approval of the Ministry Representative, at proper times, all shop and installation drawings as requested by the Contract or which the Ministry Representative may deem necessary.

In accepting the Contractor's drawings, the Ministry Representative will assume no responsibility as to the correctness of such drawings and the Ministry Representative's examination is only to ensure general compliance with the Contract.

145.13 "Ministry's Instruction" and "Contractor's Declaration"

145.13.01 "Ministry's Instruction" - "Ministry's Instruction" form (H0202) may be issued by the Ministry to the Contractor in order to:

a) describe a portion of the Work which may not be covered specifically in the Contract.

b) confirm instructions which may affect the method of carrying out the Work or payment.

c) confirm verbal agreements made with the Contractor's representative.

d) respond to the Contractor's representation on "Contractor's Declaration" form (H0203).

145.13.02 "Contractor's Declaration" - The Contractor is to issue the "Contractor's Declaration" form (H0203) to the Ministry Representative in order to:

a) make a declaration of acceptance or objection to contents of the "Ministry's Instruction".

b) formally present the Contractor's opinions, problems or requirements for clarification.

145.14 Protection and Restoration of the Work - The Contractor shall protect all Work and surrounding infrastructure from injury or damage until the issuance of the Completion Certificate.

If the Contractor fails to prevent any injury or damage

to the Work, the Contractor shall rebuild, repair, and restore at its expense any portion of the Work so injured or damaged.

145.15 Materials Supplied by the Contractor - The Contractor shall provide all materials necessary for the orderly completion of the Work with the exception of the materials supplied by the Ministry as listed in the Contract.

Material supplied by the Contractor shall not have been previously used and shall be in new condition.

All materials incorporated into the Work shall conform to the requirements of the Contract unless previously approved otherwise by the Ministry Manager.

Materials supplied by the Contractor must be approved prior to incorporation in the work.

145.15.01 Inspection of Materials - The Contractor must provide manufacturer certification or milling slips for the Ministry's use, and allow 2 weeks for the Ministry Representative to test the materials. Such certificates or mill slips must demonstrate the material properties of the specific batch of materials provided to the Project and must demonstrate that the material complies with the requirements of the Contract.

The Ministry may also perform random inspection of the materials.

The Contractor shall notify the Ministry Representative of the sources of materials supplied by the Contractor, and such notification shall be given sufficiently far in advance of delivery to enable the Ministry Representative to make inspection of the material at the source. Insufficient notice will be cause for the Ministry to reject the materials without compensation.

The Contractor will be charged for all travel time, board and lodging costs incurred by the Ministry or its designate, required to inspect materials components if the source of the material is outside the Province and incorporated into the Work.

The Ministry will offset any costs for inspecting the materials from any payments made to the Contractor.

145.15.02 Samples and Testing - The Contractor shall, as required, submit to the Ministry Representative, for approval, samples of the materials proposed for use in the work. The approval or acceptance of samples of material shall not preclude the rejection of any material upon discovery of defects or weaknesses previous to acceptance of the complete work.

Any testing done on samples of materials provided to the Ministry Representative will be performed at the Ministry's expense, except where such testing establishes that the material does not comply with the Contract. In such cases, the Ministry may offset the cost of testing from any payments due the Contractor.

145.15.03 Alternative Materials - Brands, species and manufacturer's names specified are approved for use in the performance of the Work. No alternative materials will be considered unless approved in advance by the Ministry Manager in writing.

Savings as a result of using alternate materials will be evaluated using DBSS 125 Value Engineering.

145.16 Materials Supplied by the Ministry - Ministry supplied materials will be arranged by purchase orders for delivery by the Purchasing Commission on instructions from the Contractor notifying the Ministry Representative.

Separate purchase orders will be issued for each class of material.

Materials ordered for a specific Site shall not be transferred to any other Site without the written consent of the Ministry Manager.

Ministry supplied and owned materials will be used only for Ministry Work.

145.16.01 Delivery Points - Each purchase order will specify points of delivery. The Contractor shall be responsible for all costs that may arise such as unloading from truck or rail car, hauling to the Site, stockpiling, dunnage, and reloading.

The costs to transport all materials to the Site will be incurred by the Contractor at its own cost and will be considered incidental and no separate payment will be made for this portion of the Works.

Should the Contractor request delivery be made to an alternative point other than the nearest specified point of delivery or in any manner of delivery other than specified by the Purchasing Commission, the additional cost for this delivery shall be charged to the Contractor's account.

145.16.02 Delivery of Materials – When the Ministry Representative advises the Contractor of the name of the supplier(s), the Contractor shall be responsible for:

a) determining requirements and ordering far enough in advance that work is not delayed.

b) acceptance of all materials; careful unloading and handling; hauling and delivery from delivery point to job site; storage and security; prompt turn around of tank or shipping cars, trucks or delivery containers.

c) prompt payment of all demurrage or rental charges on cars, freight, handling, delivery, storage or any other charges arising after the initial receipt of material at the delivery point.

d) ensuring that proper accounting is kept of all materials received.

The Contractor will obtain approval from the Ministry Representative prior to removing any materials from a Ministry storage site.

The Contractor shall not during normal working hours unload or take delivery of any material unless the Ministry Representative is present.

Where the Contractor takes delivery of materials during non-working hours, the Contractor shall immediately notify the Ministry Representative at the beginning of the next shift and allow sufficient time for the Ministry Representative to check quantity and quality before any materials are used.

After taking delivery of any materials supplied to the Contractor by the Ministry under the Contract, the Contractor shall be solely responsible for same and shall make good at the Contractor's expense any loss of or damage to such materials.

The Contractor will provide a secure yard or storage area for materials supplied by the Ministry. The Contractor will replace at its own cost any material, which cannot be accounted for in the storage yard inventory.

145.16.03 Distribution of Materials - To control distribution of Ministry supplied materials and to facilitate accounting:

a) the Contractor shall arrange for delivery of the materials only after applying for and receiving authority to do so on "Authority to Arrange for Delivery of Construction Materials" form (H0313). The requirements of the Contractor shall be indicated on Part "A" of the form. Approval from the Ministry Representative shall be indicated on Part "B" of the form. The Contractor may then arrange for delivery from the supplier. As delivery will be made only on receipt of the purchase order number and the serial number of the "Authority" form, the Contractor shall arrange with the Ministry Representative to immediately distribute appropriate copies of the "Authority" form.

Material suppliers will be required to notify both the Ministry Representative and the Contractor when materials are shipped, giving sufficient information to identify each and every shipment against an appropriate "Authority" form serial number and purchase order number.

b) The use of "Authority" form shall not in any way relieve the Contractor of any responsibility imposed under DBSS 145.16.02.

c) If the Ministry Representative refuses to authorize the delivery of materials to the job, the Ministry Representative will send the refusal, with reasons in writing, to the Contractor.

d) The Ministry may, at the completion of the Work, if materials are delivered in excess of job requirements through conditions beyond the control of the Contractor, take over the surplus material or grant the Contractor the option of purchasing the material on Site. The transaction will be completed only after the Contractor obtains for the Ministry Manager, a supplier's credit note for the total cost to the Ministry of the excess material purchased f.o.b. at the delivery point.

Upon completion of the Work, the Contractor will deliver all unused materials, which have been purchased by the Ministry, to a location as specified by the Ministry Representative.

145.17 Operation of Pavement Brooms on Highway - No mechanical pavement broom shall be operated on any highway without a rotating amber flashing light in operation. The light shall be mounted on self-propelled brooms and upon the towing vehicle of trailer type brooms.

Traffic Control is to be in accordance with the Traffic Control Manual for Work on Roadways. In addition, on extremely winding narrow roads an additional shadow vehicle shall be used ahead of the broom with a C-37 on the front of the vehicle and a flashing arrow in bar mode.

145.18 Supply and Maintain Facilities

145.18.01 Working Office – When specifically required under the Contract, the Contractor shall

provide a suitable working office with lock-up for the sole use of the Ministry. The office shall meet the requirements of Appendix A and be weatherproof, reasonably soundproof, and have ample window area and ventilation. Location of the office shall be determined by the Ministry Representative.

The Contractor shall regularly collect refuse and keep the office clean and properly maintained with heat and light. .

The office and contents shall be for the use of the Ministry for the duration of the Work and may if necessary be used concurrently with the other inspection agencies.

The Contractor shall supply a private telephone line for the sole use of the Ministry Representative. Ministry's long distance calls will be paid for by the Ministry.

The Work will include the provision of the office, furnishings, maintenance through out the Term, and removal in its entirety.

145.18.02 Testing Facility – When specifically required by the Contract, the Contractor shall provide the Ministry with a material testing facility with a minimum of 12 m^2 floor space either in the form of a small trailer or the portion of a large trailer. The testing facility shall include power, light, heat, ventilation, plus basic furniture to permit testing and preparation of test records. The facility shall be stable, reasonably quiet and lockable.

The Work will include the provision of the facility, furnishings, maintenance through out the Term, and removal in its entirety.

145.18.03 Not Used

145.18.04 Weigh Scale - The Contractor shall provide a truck weigh scale wherever measurement of materials by mass is required.

The weigh scale shall be an approved electronic weigh scale or platform type with remote balancing beam. The scale shall be of sufficient size and capacity to weigh, at a single pass, the largest, rigid-framed haulage vehicle to be used on the job and in any case shall be of not less than 13 t (tonne) capacity. The scale shall be capable of weighing all loads to within 20 kg of the true mass.

At the start of the Work, and as frequently as the Ministry Representative may deem necessary in order to ensure the accuracy, the scale shall be calibrated and checked at the Contractor's expense by an Inspector of Weights and Measures.

The remote balancing beam shall be enclosed in a sound, watertight, dust-free scale house, properly ventilated and insulated. Adequate heating must be provided when the shade temperature is below 16°C.

The scale house shall be provided with full length windows facing the scale platform, and windows shall also be provided at either end, so that those in the scalehouse can at all times see vehicles to be weighed, not only on the scale but also approaching and leaving.

The front window shall be sufficiently large and low that the whole scale platform is readily visible from the scale house. Immediately behind the balancing beam, a bench or table of sufficient length shall be provided to permit both scaleperson and checker to sit abreast in a position in which they can both write on the table and view the electronic scale or weigh beam and scale platform. Drawing SS SP145-01 shows these requirements.

The Contractor shall supply and maintain portable washroom facilities at the weigh scale site.

145.19 Weighing Materials - The Contractor shall supply a qualified scaleperson to carry out the necessary weighing.

A weighing certificate prepared in triplicate, certified by the scaleperson, will be issued for each load.

The delivery receipts will bear the signature, initials or other approved marking of the issuing scaleperson and receiving checker.

If the Contractor has an automated scale and the following conditions are being met, the Contractor shall not be required to supply a scaleperson:

a) All weighed material is being delivered to a single job site. If the Contractor is delivering to two or more job sites at the same time, the Contractor shall supply a scaleperson as detailed in the first paragraph.

b) Truck operators do not require instruction.

c) There are no overloads, tarp covering, spillage difficulties, or any safety problems.

d) The scale platform is maintained by the Contractor to be clean and free from loose material at all times.

All weighed vehicles must be tared at least once a day.

145.20 Weather - No work shall be undertaken by the Contractor when, in the opinion of the Quality Manager, the weather is unsuitable or unfavourable for that particular class of work.

145.21 Layout of Work and Calculation of Quantities- A number of key reference points of control line (primary) for the establishment of the design control line may be provided by the Ministry to aid the Contractor to lay out the Work. The Contractor shall assume full responsibility for the alignment, dimensions and elevations of each and every part of the work and their mutual agreement.

145.21.01 Not Used

145.21.02 Contractor's Responsibilities - The Contractor shall perform all layout of the work and all other survey work required to execute the Contract and construct the works in accordance with the Contract.

The Contractor shall be responsible for establishing all control points and/or lines, all slope stakes, the establishment of line and grades for subgrade and the various granular aggregate layers, layout by line and grade of all structures, culverts, and underground utilities, and shall perform all other layout and measurement necessary for the proper execution of the Contract.

The staking of all works shall be of a sufficient accuracy and frequency for the Ministry to carry out its quality management program.

On request of the Ministry Representative, the Contractor shall submit documentation to verify the accuracy of the layout work.

The Contractor shall be satisfied, before commencing any Work, as to the meaning, intent and accuracy of any control points, control lines and benchmarks established by the Ministry.

Should the Contractor discover or suspect any errors in any control points, control lines, benchmarks, and data provided by the Ministry, the Contractor shall at once discontinue the affected work until such errors are investigated by the Ministry Representative and, if necessary, rectified.

Unless the Design requires a tighter tolerance, all Work shall be laid out and constructed to within ±50 mm of its Design location.

Unless the Design requires a tighter tolerance, the Work shall be so laid out and constructed that the location of substructure elements and anchor bolt groups shall not differ from the true positions by more than the following amounts:

a) Substructure elements - 12 mm

b) Anchor bolt groups - 6 mm.

c) The elevation of bridge seats shall not differ from true elevations by more than 3 mm.

Any deviation beyond the above mentioned tolerances shall be satisfactorily corrected by the Contractor at the Contractor's expense.

145.22 Access to Site - The Contractor shall provide access to the Site as may be necessary.

The Contractor shall make arrangements with the property owners adjacent to and around the terminal site for the use of access and for the use of any storage areas the Contractor may require. The Contractor shall acquire such access and storage rights at the Contractor’s expense.

In the event of roads being temporarily closed to the travelling public, the Contractor shall, at the Contractor's expense, provide, erect and maintain all requisite barriers, fences or other proper protection and must provide, keep and maintain such patrollers, lights and danger signals as may be necessary, or as may be required by the Designer or Ministry Representative to ensure safety to the public and to the workers.

The Contractor, during the Contractor's operations, must also provide and maintain reasonable road access and egress to property fronting along or in the vicinity of the work under contract, unless other means of road access exists. The Ministry Representative will be the sole judge as to what may be deemed reasonable road access.

145.23 Accommodation of Traffic - The Contractor shall cause as little inconvenience as possible to the travelling public during the Contractor's operations and shall erect and maintain proper and adequate barricades, traffic signs, lights and other traffic control devices as may be considered necessary, in the opinion of the Designer or Ministry Representative, for the safety of both workers and public traffic.

All barricades, traffic signs, lights and other control devices shall be provided at the Contractor's expense, and shall be erected in accordance with, and otherwise conform to the standards as set out in the Ministry's "Traffic Control Manual for Work on Roadways",

Contract and DB Standard Specifications.

145.23.01 Maintenance of Traffic - The Contractor shall comply with the requirements of the Transportation Act and amendments thereto and such other Acts, By-Laws or Regulations as are in force for the governing and regulating of traffic or use of any road upon or over which it is necessary to do work or transport materials. The Contractor must arrange to keep roads open to traffic at all times and must cause as little inconvenience as possible to the travelling public, except as permitted under the Contract.

145.23.02 Road Closure Notification - Road closures shall not be permitted until the Contractor requests and receives written permission from the appropriate authority in the Ministry. This request shall be submitted in writing not less than seven days in advance of the closure.

Upon receiving written permission for closure, the Contractor shall notify the local police, fire, ambulance, municipality, school board, and public transit at least 72 hours in advance of the closure.

The Contractor shall notify the public by local radio and newspaper at least 24 hours in advance of the closure.

145.24 Not Used

145.24.01 Not Used

145.25 Use of Explosives - In using explosives the Contractor shall exercise the utmost care so as not to endanger life or property and, whenever directed, the number and size of the charges shall be reduced. Before doing any blasting the Contractor shall ascertain if there are in the neighbourhood any railway, power or other lines, water or other pipes, etc., which are likely to be in danger of being damaged. The Contractor shall give to the owner of such lines, pipes or other works timely notice of the intention to blast.

Whenever possible electric batteries shall be employed for simultaneous blasting.

The storage of explosives shall be in accordance with the regulations of the Canada Explosives Act and with any requirements of the representative of the Explosives Regulatory Division within the Mineral Technology Branch of the Minerals and Metals Sector of the Federal Department of the Natural Resources Canada.

145.26 Work in Ministry Pits or Quarries - When operating in a Ministry pit or quarry where the Ministry retains the role of Pit Manager, the Contractor shall comply with all provisions of the Mines Act, the Health, Safety and Reclamation Code for Mines in British Columbia, and/or the WCB Occupational Health & Safety Regulation, as applicable to the work undertaken.

The Contractor shall appoint a properly qualified supervisor or shift boss in any Ministry pit or quarry where work is undertaken, in compliance with Section 26 of the Mines Act, and ensure that the supervisor or shift boss is duly qualified certified in accordance with Section 1.13 of the Health, Safety and Reclamation Code for Mines in British Columbia.

When the Contractor is operating in a Ministry pit or quarry, the Contractor shall utilize the pit or quarry in accordance with the Ministry Pit Development Plan or obtain written approval from the Ministry Project Manager prior to deviating from the plan.

When investigating private aggregate sources, the Contractor shall ensure that all permits are in place allowing the Contractor to purchase and remove the aggregate from the property of the Owner.

Availability and use of the listed reserves or any other private aggregate source must be arranged by the Contractor. Prior to the removal of any aggregate material from an aggregate source, the Contractor shall provide a written notice from the owner authorizing the Contractor to remove aggregate material for use in the work.

Ministry pits or quarries may be used without charge, provided they are utilized, worked and maintained in accordance with the DB Standard Specifications and the Ministry's development plans. Prior to the removal of any aggregate material, the Contractor shall obtain written approval from the local District Manager Transportation. No warranty is given or implied that the source of aggregate material from the Ministry pits or quarries will meet requirements or requisite quantities.

The Contractor shall contact the Road and Bridge Maintenance Contractor a minimum of two (2) weeks before the date the Contractor anticipates the use of any Ministry pits or quarries.

The Contractor and the Road and Bridge Maintenance Contractor shall co-ordinate their activities and schedule usage of the pit among themselves and any other affected parties.

The following listed Variances to the Health, Safety and Reclamation code for Mines in British Columbia will apply only to pits and quarry operations which fall under Ministry ownership.

The Chief Inspector of Mines has provided the following clarifications to the Ministry in a letter dated August 9, 1999:

Application for Continuation of Variance to Mine Code Section 1.5.1 - Hours of Employment

"Pursuant to section 13 of the Mines Act, R.S.B.C. 1996. c293, the assigned Ministry of Transportation (MoT) Pit Manager may permit the employment of persons in a sand and gravel pit or quarry (Mine) for up to 12 hours in a 24 hour period.

This variance applies only to pits managed by the MoT (Ministry of Transportation).

The Pit Manager shall communicate this variance to all affected workers, whether employees of the Ministry of Transportation, its contractors or sub-contractors.

This variance is subject to review within five years as provided by section 13(4) of the Mines Act.

A copy of this variance is available to any MoT regional or district Occupational Health and Safety Committees which may be involved in sand and gravel pit or quarry operations."

Section 4.9.1 – Health, Safety and Reclamation Code for Mines in British Columbia (Code)

"Due to the short period of time that some haul trucks spend at your various pits and quarries, i.e. for loading purposes only, I am hereby enabling a Health and Safety Inspector of the Mines Branch to vary the requirement for automatic back-up alarms on transient haul trucks at the discretion of the Inspector. If, in the opinion of the Inspector, the time spent by the truck at the pit or quarry is of short duration, and the required routing of the truck for loading purposes is such that no hazards are created by such a routing, then an automatic back-up alarm need not be required and the use of any audio warning device will be accepted while in reverse."

"You are reminded that the above refers only to transient haul trucks and not to trucks used for production purposes; neither does it apply to any other vehicle in excess of 7,000 kg gross vehicle weight."

Section 4.9.5 - Health, Safety and Reclamation Code for Mines in British Columbia (Code)

As a haul truck driver's visibility of the area surrounding the truck tends to improve as the size of the truck decreases, I am prepared to allow a variance to the above section of the Code as follows:

"Provided that any haul truck used for production purposes at a gravel pit or quarry does not exceed 45 tonnes gross vehicle weight, there will be no requirement for the attachment of a whip antenna fitted with a flag and lamp, or a flashing light mounted on the cab of non-production vehicles."

145.26.01 Available Ministry Sources - The Contractor, by entering an available source, will be deemed to have accepted all obligations, risks and costs for the production of aggregate meeting the contract requirements, just as if the pit or quarry was private. No warranty is given that the source will meet the quality and quantity requirements. The Contractor, without any additional compensation, shall provide whatever selection, screening, crushing, mixing, blending or other processing of any kind may be required to produce aggregate meeting all requirements for its intended use, including but not limited to:

- specified aggregate gradation,
- fracture count,
- aggregate quality requirements such as but not limited to those set out in DBSS 202.04 and Table 501-D of DBSS 501 as applicable, and
- the specified properties of any products into which the aggregate is to be incorporated.

The Contractor shall comply with the Ministry pit development plan and the Ministry Representative's directions regarding the location of material extraction within the pit by the Contractor.

The Contractor's use of Ministry pits is limited to the extraction of previously unexcavated material from the bank, for the supply of materials for use on this Contract.

The Ministry's Road and Bridge Maintenance Contractor has certain rights and obligations with respect to some Ministry pits, and the Contractor shall coordinate its activities with those of that contractor, where applicable, and with any other users, to the satisfaction of the Ministry Representative and the Ministry's District Manager, Transportation.

145.27 Highway Right-of-Way - The Contractor shall confine its operations to the highway right-of-way and Licences to Construct as shown on the Ministry's Drawings. Any land, property or water lot outside these boundaries required for construction yards, access roads, disposal or storage areas, or other operations of the Contractor, shall be acquired by the Contractor at no additional expense to the Ministry. The Contractor shall make arrangements for the use and compensation of such land with the titled land owner.

Work within the Site shall be confined to that necessary to construct the Project to the lines and grades specified. Excavation outside the design lines and grades to extract aggregates or for other purposes is prohibited unless authorized in advance by the Ministry Representative.

145.27.02 Disposal Site Outside the Right-of-Way - Waste, or other materials specified or designated by the Ministry's Representative to be disposed of, such as, but not limited to clearing and grubbing materials, fences, old culverts, old concrete and asphalt, etc., shall be removed from the Site to a disposal site provided by the Contractor. All disposal sites shall be the responsibility of the Contractor.

The Contractor shall ensure that the disposal site complies with all Federal, Provincial, Municipal and Regional by-laws and regulations prior to disposing of the material. Any fees incurred shall be paid by the Contractor. The Contractor must provide documented proof to the Ministry Representative for the use of the selected disposal site.

In addition, the Contractor shall ensure that the disposal site complies with all municipal and regional government by-laws and regulations and, when relying upon any exemption permitted under section 2(d) of the *Agricultural Land Reserve Permit Regulation*, shall provide a written confirmation from the Regional District of their agreement that no permit is required.

145.27.03 Access Roads to Private and Public Lands - The Contractor shall not block or impede the use of any private access roads or driveways during the Work of this Contract without the consent of the Ministry Representative.

The Contractor shall schedule any anticipated closure of an access with the property owner to minimize impact on the owner.

145.27.04 Private Land Reserves Borrow material shall not be obtained from land outside the right-of-way, without prior approval from the titled land owner. In addition, excavation or deposition on lands located within an Agriculture Land Reserve require the approval from the B.C. Agricultural Land Commission and issuance of a valid permit under the Soil Conservation Act.

145.28 Protection of Land Subdivision Monuments, Roads and Property

145.28.01 Protection of Subdivision Monuments - The Contractor shall protect, preserve, and keep uncovered all land subdivision monuments or property marks during its construction activities which may cause disturbance to them. At points where construction activities will cover or destroy any land subdivision monuments or property marks, the Contractor shall restore them afterwards.

145.28.02 Protection of Survey Monuments - The Contractor shall be responsible for the preservation during construction of all geodetic benchmarks, survey monuments and property markers on the right-of-way. The Contractor shall use, at no additional expense to the Ministry, a British Columbia Land Surveyor to replace any survey monuments destroyed or damaged as a result of the Contractor's negligence.

145.28.03 Protection of Roads - The Contractor shall ensure all equipment working on or hauling material on to and from the Site does not damage or deposit material onto any part of an existing roadway. Materials spilled onto the public roadways or driveways opened to public traffic shall be cleaned up immediately. The Contractor has the full responsibility to repair any damage to existing highways, local roads and driveways caused by its construction equipment and/or operations.

145.28.04 Protection of Railway Property - The Contractor shall be wholly responsible for any damage to railway trackage or property on the railway company's right-of-way resulting from highway construction operations. It shall be the Contractor's responsibility that all workers and persons employed by the Contractor or the Contractor's agents, or under the Contractor's control, shall use due care that no person or property is injured, and that no rights are infringed in performing the work. The Contractor shall compensate injured parties.

The Contractor shall be responsible for:

- traffic control persons or other personnel required for the protection of the railway trackage

- re-routing or repairing any telegraph or telephone lines during the course of operations.

The Contractor shall, to the satisfaction of the Ministry Representative, restore and clean all slopes between the highway and the railway track together with any fences, ditches and drains disturbed by the work.

The Contractor shall give advance notice to the railway company prior to carrying out any work on the railway right-of-way and shall not proceed without specific coordination with the railway company and its schedule, and notice to the Ministry Representative.

145.28.05 Not Used

145.28.06 Protection and Restoration of Property and Environment - The Contractor shall not enter upon private property for any purpose without first obtaining permission of the owner of that property and shall be responsible for the preservation of public property along and adjacent to the Work, and shall use every reasonable precaution necessary to prevent damage or injury thereto.

The Contractor shall be liable for all damages caused by fire started by the Contractor and shall, under no circumstances start fires without first securing the required permits and approval of authorities having jurisdiction even though it may be ordered or required to do such burning.

The Contractor shall take special care when burning brush, stumps or rubbish not to damages any standing trees, shrubs or other property and to use when practicable smokeless burning methods.

145.29 Haul Routes and Overloads - The Contractor shall be responsible for ascertaining that the Contractor's haul routes are acceptable to the authority having jurisdiction for highway loading.

Overweight or oversize vehicles are not permitted to use existing highway as a haul road for the hauling of construction materials, notwithstanding Section 7.01 of the Commercial Transport Regulation 1980 Consolidation, as amended from time to time.

Off-highway equipment may be permitted to cross the existing highway only at specific locations as approved by the Ministry Representative.

Once the crushed granular base, surfacing or pavement work has commenced on any portion of the new, reconstructed or relocated highway, no overloads will be permitted on that portion unless written approval is given by the Ministry Representative.

145.30 Health and Safety

145.30.01 Occupational Health and Safety Program - The Contractor shall provide an Occupational Health and Safety Program and have it in place prior to the commencement of any Work on Site.

Any delays in commencement of the Work related to acceptance of the Contract and site specifics of the Contractor's Occupational Health and Safety Program or interruptions of the Work related to the implementation of the Occupational Health and Safety Program are not Reimbursable Delays unless otherwise agreed to in writing by the Ministry Representative.

145.30.02 Worksite Hazards - The Contractor has the responsibility to identify worksite hazards and shall develop operational occupational safety policies and procedures and plans which are specific to the Work to ensure the safety of all persons at the Site and the travelling public passing through the Site.

When requested by the Ministry Representative, the Contractor shall provide copies of any safety policies, procedures or plans verifying steps have been taken to make all concerned fully aware of all hazards or unsafe conditions in accordance to DBSS 135 - Construction Site Safety.

The Contractor shall provide for the safety, protection and convenience of the general public and residents along the highway.

The Contractor shall erect and maintain signs, fences and barricades at the ends of the bridges before they are opened to traffic and at other dangerous areas of the Project, as necessary to prevent access of vehicles and pedestrians to the unfinished bridge and other dangerous areas.

145.30.03 Equipment - All trucks and other equipment used by the Contractor whether owned or rented for use on the Project shall meet manufacture's specifications for its purpose.

All trucks and other equipment rented by the Contractor for use on the Project shall when practicable be qualified local trucks obtained from local residents.

For the purposes of this Section, qualified local truck means a truck that meets all the following:

a) a truck that has Motor Carrier approval.

b) where the Contractor has a collective agreement with its employees the truck operator is a member of, or has the permission of the appropriate bargaining unit.

c) the truck fulfils, in the opinion of the Ministry Representative, the specialized needs of the Contractor.

d) a truck registered on the Ministry's Hired Equipment list for the geographic area encompassing the Site to in adjacent areas.

Hire of trucks and other equipment paid on a production rate basis constitutes a Subcontract.

145.30.04 Audible Back Up Warning Device - All haul vehicles shall be equipped with an automatic audible warning device which will activate whenever the vehicle is backing up. The warning device shall be clearly audible above the ambient noise level at a minimum distance of 6 m from the back of the vehicle

145.31 Faulty Work - If there is evidence of any fault, defect, or injury from any cause whatever, which may adversely affect the strength, durability, safety, maintenance, or appearance of any section of the work, the Contractor shall, at the Contractor's expense, satisfactorily correct such faults or, if required, shall replace as much of said section as the Ministry Representative may deem necessary, even to the extent of replacing the entire section. The fact that the inspector in charge may have previously overlooked such defective work shall not constitute an acceptance of any part of it.

145.32 Temporary Excavation - Where temporary excavations are undertaken in the course of the Work, the Contractor shall be responsible for maintaining the stability of adjacent ground and all structures and other works thereon.

145.33 General Sign Requirements

145.33.01 Contractor Identification Signs - The Contractor may supply and install contractor identification signs for the Term. The Contractor shall maintain and remove such signs and all costs of supply, installation, maintenance and removal shall be the responsibility of the Contractor.

The contractor identification sign shall display only the Contractor's company name and beneath it the word "Information" followed by a telephone number which directly connects any caller to the Contractor's personnel who are familiar with, and are currently active on the Project.

The sign will not be reflectorized, and the background colour will not be either red, yellow or orange. The sign dimensions will not exceed 2,400 mm horizontal and 1,200 mm vertical, and the text size (height) will not be less than 150 mm.

The contractor identification sign will be erected only in conjunction with the C-35 project identification sign, and will not be in advance of or at the same location as the C-35 sign. Where there is no C-35 sign, no contractor identification sign will be permitted.

No contractor identification sign may be installed without prior approval of both the sign and the intended installation and location by the Ministry Representative who, in determining whether or not to grant such approval, will not necessarily be limited to the requirements set out above.

145.33.02 Prime Contractor Sign - When designated as the Prime Contractor, the Contractor shall provide “Prime Contractor” sign as specified in DBSS 135 Construction Site Safety.

145.33.03 Safety Apparel and Traffic Control Signs – All traffic control signs shall meet current Ministry specifications. The Ministry *Specifications for Standard Highway Sign Materials, Fabrication and Supply* are located on our Ministry web page at:

www.th.gov.bc.ca/publications/eng_publications/ geomet/geometsigns.htm

The safety apparel and traffic control retro-reflectivity signs and devices referenced in the Traffic Control Manual for Work on Roadways (TCM) are being revised as per Technical Circular T-09/05.

http://www.th.gov.bc.ca/publications/Circulars/Current/T_Circ/2005/t09-05.pdf.

Notwithstanding the 3-year phase-in period described in the circular, the new standards shall apply to all Work under this Contract.

APPENDIX A

Ministry Office Requirements

REQUIREMENTS FOR OFFICE TYPES			
Office Type	**A**	**B**	**C**
Lockable desk/chair	1	2	3
Typist desk/chair	-	-	1
Drafting table 1.2 m x 2.4 m /stool	1	1	1
Reference table 1.2 m^2	-	1	3
Conference table 1.2 m x 2.4 m	-	-	1
Additional chairs	2	2	9
Plan case or drawing storage unit	1	1	2
Filing cabinet (# drawers)	1 - 2 drawers	1 - 4 drawers	3 - 4 drawers
Shelving bookcase m^2	2	5	20
Fire Extinguisher for Class A, B and C fires	1 - 5 kg	1 - 5 kg	2 - 5 kg
Supply Activated Telephones and Lines	One (1)	One (1)	Three (3) on two (2) lines with external bell and switch
Toilet	On site	On site	In washroom, hot and cold running potable water and wash basin
First Aid Kit	On site	#1 Unit Kit	#1 Unit Kit
Floor Areas - m^2			
a) Entrance	-	-	5
b) Ministry Representative's office	12	12	12
c) Additional Staff office	-	8	20
d) Wash room	-	-	As required
e) Store room	-	-	5

SECTION 165

PROTECTION OF THE ENVIRONMENT

165.01 General - This Section covers the general and specific provisions for the protection of the environment, in cooperation with any Federal and/or Provincial Environment officers or an Environmental Monitor, as designated in the Contract.

165.01.01 Intent of Specifications - These Specifications give the Ministry the right to exercise control over environmental aspects of the work. The Contractor shall adhere to specific instructions if the work may result in an adverse impact on the environment.

The Ministry will determine the significance of environmental impacts in consultation with Environmental Agencies. The Ministry reserves the right of approval over the general methods employed by the Contractor in the performance of the work, but only insofar as they may affect relations with Environmental Agencies and the protection of aquatic and terrestrial resources, the health and safety of public, and protection of socio-community resources and features.

These specifications are for the protection of the environment, and shall be given such interpretation as will secure this intent.

165.01.02 Definitions - For the purposes of this Section, the following general terms are defined as follows:

“Compensation” refers to monetary payment or replacement in kind for environmental losses resulting from a development project, as well as the construction activities for the re-creation of lost or damaged habitat.

“Deleterious Substance or Material” is defined as a substance harmful to fish or fish habitat (Canada Fisheries Act, Section 34.1).

"Designated" shall mean designated in the Contract or by the Ministry Representative.

“Environment” refers to the physical, biological, social, spiritual and cultural components that are interrelated and affect the growth and development of living organisms. The term “environment” in these specifications shall include socio-community issues and resources.

"Environmental Agencies" shall mean the appropriate regulating branches of the Federal and Provincial agencies responsible for the management and protection of the Environment and human resources. A partial listing of these Environmental Agencies and their areas of concern is given in the joint publication "Land Development Guidelines for the Protection of Aquatic Habitat", Fisheries and Oceans Canada and Ministry of Environment, Lands and Parks (1992), hereafter referred to as the "Land Development Guidelines".

The "Environmental Monitor" shall mean an agent hired by the Contractor's in compliance with the environmental protection aspects of Standard Specifications, the Contract, permits and approvals, and to advise the Contractor and Ministry Representative on environmental problems. An Environmental Monitor will be used when and where required, and will report to the Contractor.

“Environmental Approval” is defined as the written authority issued to a person or company by a government agency that allows the person or company to do something that otherwise may not be permitted by law or which is not defined in law. An environmental approval is a broad, generic term referring to informal or formal authorization for actions that may have an adverse effect on the environment, such as (i) undertaking an activity (e.g., authorization from fishery agencies to proceed with work within the wetted perimeter of a fish-bearing stream); or (ii) discharging some form of material (e.g., approval under the British Columbia Waste Management Act to introduce waste into the environment or the storage of special waste for a period of 12 months or less). The term “approval” can include related forms of authorization such as permits and licenses.

“Environmentally Sensitive Areas” shall mean areas requiring special management and attention to protect resources, habitat or species (which includes and is not limited to water quality, identified sensitive areas, fish and fish habitat, vegetation, rare and endangered flora/fauna, landscaping and visual aesthetics, soil conservation (including dust control), air quality and archaeological, heritage and cultural resources).

“Environmental Timing Window” shall mean any period suitable for environmentally sensitive construction work as designated by an Environmental Agency or other federal, provincial, municipal or local agency. This window represents the period that the natural or human environment is likely less susceptible to adverse impacts. A common example of an Environmental Timing Window on highway construction projects is a Fishery Timing Window. Timing windows may also be imposed for the protection of breeding birds, retention of significant recreational activities, maintenance of critical traffic patterns, etc.

“Fishery Timing Window” refers to the time period(s) of reduced risk for important commercial, sport, and resident fish species, based on their life histories. The Fishery

Timing Window is the time of year during which there are no fish eggs or alevins present in the substrates of local watercourses, and the period when fish migration (juvenile out-migration and adult spawning in-migration) is not occurring. This is generally the preferred period for (i) instream work or (ii) work adjacent to or over top of fish-bearing streams with the potential to create adverse impacts on fish or fish habitat.

"Habitats" are defined as those parts of the environment on which terrestrial and/or aquatic species depend, directly or indirectly, in order to carry out their life processes.

"Fisheries Sensitive Zone" is defined as the instream aquatic habitats, as well as out-of-stream supporting habitat features such as side channels, wetlands, and vegetated riparian areas adjacent to these features.

“Impact” is defined as an alteration, either positive or negative, to the environment brought about as a direct or indirect result of a highway project, including construction, operation and maintenance work (e.g., the consequence of a highway-related activity interacting with its surroundings).

“Mitigation” refers to a procedure or an action designed to avoid, reduce or control the severity, magnitude, duration and/or frequency of environmental impacts of a project through design alternatives, scheduling or other means.

“Permit” refers to a formal authorization, typically granted to proponents by an Environmental Agency, for discharging agreed upon quantities and types of regulated substances such as pollutants or waste. For example, under Section 8 of the British Columbia Waste Management Act, a permit may be issued to introduce waste into the environment or to store special waste subject to environmental protection requirements which are deemed advisable. The permit can specify the procedures or requirements respecting the handling, treatment, transportation, discharge or storage of waste that the holder of the permit must fulfill. The term “permit” can be defined by applicable legislation. The term “permit” is synonymous with the term “license”, and the two terms are often used interchangeably.

“Riparian Area” is defined as the land adjacent to the normal high water level in a stream, river, lake or pond and extending to the portion of land that is directly influenced by the presence of adjacent ponded or channeled water, or a groundwater zone fed by surface water bodies (e.g., zone in which rooted vegetation is influenced). Riparian areas typically contain important vegetation resources which provide several critical functions for the survival of fish (e.g., protection from predators, shade for temperature regulation of the watercourse, and sources of food).

“Sedimentation” is defined as the deposition of material carried in water, usually as a result of a reduction in water velocity below the point at which material can be transported.

"Watercourses" shall apply to all bodies of water including streams, rivers, canals, ditches, lakes, ponds, and wetlands.

“Wetted Perimeter” is defined by Station number and/or as highlighted on the plan drawings.

165.01.03 General Restraints for Watercourses and Groundwater - The following general restraints in regard to the protection of any watercourse shall apply:

a) The Contractor shall place and/or dispose of all organic material, refuse, ash, petroleum products and other deleterious materials so as not to directly or indirectly pollute any watercourse or groundwater. The placement and disposal of all such products and materials shall be done in an environmentally acceptable manner.

b) Except as required by the Contract, all inorganic material shall be placed and/or disposed of in a manner that does not obstruct or unduly disturb any permanent or seasonal watercourse. Any such obstruction or disturbance shall be restored to the original drainage pattern. Any removal of inorganic material from a watercourse shall be done in a manner that minimizes adverse impacts.

c) All activities within the wetted perimeter of any watercourse shall be kept to an absolute minimum. Machinery and equipment shall not be operated within the wetted perimeter of any watercourse other than under the terms of the Authorization and permits issued by the Environmental Agencies. An Environmental Monitor shall be present during any work within the wetted perimeter of any watercourse on the Project.

The preceding requirements, **a)** through **c)**, are considered incidental to the price bid for work under Contract and no other compensation will be made.

165.01.04 Designated Streams and Designated Environmentally Sensitive Areas - Any watercourse or environmentally sensitive area designated in the Contract is subject to all of the restraints of this Section.

165.02 Planning and Scheduling - The Contractor shall carefully plan and schedule construction activities in a manner that ensures the avoidance or absolute minimization of environmental damage. The Contractor shall be familiar with and be able to identify those areas and times which present environmental problems and shall

prepare schedules and work methods accordingly. The Contractor shall forward a copy of each of the schedules to the Ministry Representative at least one week in advance of commencement of each of these operations, unless otherwise specified in the contract.

The Contractor, shall provide an **Environmental Management Plan** (EMP) which describes in detail the approach to be taken in addressing environmental issues associated with the Project and the correlation of the EMP to the project schedule. The EMP will be structured according to the following format and will include, but not be limited to, the following:

- demonstrate understanding of the specific environmental issues involved with the Project,
- indicate an understanding of Ministry/Contractor responsibilities,
- include a sediment and drainage management plan,
- identify all the tasks required and clearly identify the duration and sequence of each task leading to the receipt of agency approvals, and
- indicate its linkage to the project schedule.

The EMP will clearly indicate how the Project will be undertaken to avoid negative impacts pertaining to, but not limited to, the following resources:

- water quality (including all surface and sub-surface sources),
- identified sensitive areas,
- fish and fish habitat,
- wildlife,
- vegetation (including riparian plant communities),
- rare and endangered flora/fauna,
- landscaping and visual aesthetics,
- soil conservation (including site stability, dust control),
- air quality, and
- archaeological, heritage, and cultural resources.

The Contractor, in consultation with the Ministry Representative, Environmental Monitor and the Environmental Agencies, shall be responsible for determining the conditions under which the work must be carried out in Environmentally Sensitive Areas, and for making all required on-site examinations and examinations of documents supplied and referenced by the Ministry in order to fully comprehend the environmental aspects of the work required.

165.02.01 Environmental Legislation and Regulatory Requirements - The Contractor shall observe and comply with all federal, provincial, municipal and local laws and regulations which seek to ensure that construction work does not adversely affect the environment or social-community resources.

In the event of conflict between the requirements set out in these Specifications and quality control laws, statutes, regulations and ordinances of federal, provincial, municipal or local agencies, the more restrictive requirements or regulations shall apply. A violation of the environmental laws and regulations reported to the Ministry by the responsible agencies may result in the issuance of a non-conformance report. Should the situation warrant more stringent measures, it may also result in the issuance of a stop-work order until the violation is corrected. The Contractor shall have no recourse for reimbursement due to delays or alterations to construction activities arising from such violations or the correction of such violations.

165.02.02 Environmental Procedures - The Environmental Agencies or the Ministry Representative may require one or more detailed sets of environmental procedures for any work in and around Environmentally Sensitive Areas, such as critical fish habitat. When a set of environmental procedures is required, the Contractor shall prepare and submit the procedures to the Ministry Representative and the Environmental Agencies prior to undertaking work in these areas. These detailed procedures shall supplement the EMP. The Contractor shall forward a copy of the approved set of procedures to the Environmental Agencies and the Ministry Representative, prior to undertaking the work. The environmental procedures shall contain the following items:

a) Existing environmental conditions. Identification of the environmental resources (e.g. fish species and habitat) in the area of the proposed work.

b) Description of work proposed in the Environmentally Sensitive Area. Summary of the proposed work, equipment to be used, schedule of activities, and location.

c) Environmental protection measures. Statement concerning the protective measures that will be used to protect environmental resources (e.g., species and habitat) from each anticipated adverse impact.

d) Contingency plan. Description of alternative or backup plan in the event of an environmental emergency or failure of any of the protective measures.

e) Environmental monitoring requirements. Indication of any specific or unique environmental monitoring requirements to ensure compliance with environmental specifications and proper implementation of the environmental procedures.

The Contractor's Environmental Monitor shall contact the Fisheries and Oceans Canada (F&OC) and the Ministry of

Environment for specific recommendations about the preparation of the environmental procedures. The environmental procedures must be approved by the Ministry Representative and the Environmental Agencies prior to commencement of work in Environmentally Sensitive Areas.

165.02.03 Project Orientation - The Contractor and their the Environmental Monitor shall arrange regular meetings with the Ministry Representative,. Other interested parties may be asked to attend (e.g., Agency Representatives), whose representation is necessary because of the specific type or location of work being undertaken. The purpose of such regular meetings will be to outline the schedule of upcoming construction and proposed activities, and to review the activities of the previous week. Such regular meetings do not relieve the Contractor from attending or arranging other types of meetings as required due to the nature or extent of the work being done.

The Contractor shall arrange and conduct such regular meetings during construction, subject to the following conditions:

a) Meetings shall be held on a weekly basis when construction activities are to proceed during an Environmental Timing Window, during a period when work is to proceed in an Environmentally Sensitive Area, or during a period when the proposed construction activity elsewhere on a project is to proceed for a duration greater than two weeks and has the potential to cause adverse environmental impacts.

b) During periods other than that specified above, regular environmental meetings shall be held every two weeks, or as otherwise agreed to by the Ministry Representative.

c) In the event that the Environmental Monitor or the Ministry Representative deem the Contractor's compliance regarding environmental requirements to be inadequate at any time during the Project, the frequency of the weekly or biweekly regular meetings shall be increased accordingly until the Contractor's Environmental Monitor and the Ministry Representative are satisfied with the compliance rate.

d) The Contractor's Environmental Monitor will record highlights from the meetings (e.g., summary of major discussion items and key action items) and forward a copy to the Contractor, as well as include this material in the Project Environmental Progress Reports.

e) During the maintenance phase of the Project, the Contractor shall conduct regular environmental meetings on a quarterly basis, unless operations are to be conducted within Environmentally Sensitive Areas that pose potential environmental concerns. In the event that operations are to be conducted within such areas, the Contractor and Environmental Monitor shall conduct weekly environmental meetings for the duration of the maintenance work in such areas.

The Contractor is responsible for ensuring that the foreman, operators, and work crews (including any subcontractors) understand the specific environmental issues on the work site and their responsibilities under this Section. The Contractor's Environmental Monitor will give all construction personnel a brief environmental orientation prior to such personnel commencing work in any Environmentally Sensitive Area on the Project. The Contractor's Environmental Monitor will ensure that construction personnel are familiar with the environmental requirements and acceptable construction practices on the Project. The Contractor shall support the Contractor's Environmental Monitor's role in this regard.

165.02.04 Activity Within Designated Watercourses and Environmentally Sensitive Areas - All activities that are conducted within the wetted perimeter of any designated watercourse or within other designated Environmentally Sensitive Areas must conform to the timing restrictions stated in the Contract, and to the Environmental Timing Windows given in the authorizations/approvals of the Environmental Agencies. Construction work must be scheduled so that all operations affecting the Fisheries Sensitive Zone or other designated Environmentally Sensitive Areas can be completed within these time periods. Prior to the commencement of any construction activities, all work in these areas must be approved by the Environmental Agencies.

The Contractor must advise the Ministry Representative in advance of any construction-related activity in designated areas. Construction activities within designated watercourses or other Environmentally Sensitive Areas shall not commence without the attendance and support of the Environmental Monitor.

165.02.05 Inclement Weather - The Contractor shall cease operations, modify construction methods, or relocate to an alternative site within the project area during periods of inclement weather to avoid siltation of designated watercourses

165.02.06 Work Stoppage - Prior to initiating work stoppages, the Contractor shall initiate and complete any mitigative and environmental protection measures required to safeguard the environment and the Project during work stoppages.

The Contractor shall consult with the Ministry Representative of impending shutdowns to enable the Ministry Representative, Contractor, and the

Environmental Monitor to examine the project. This notice shall be sufficient to permit the Contractor adequate time to install all necessary additional environmental mitigation measures.

The Contractor shall ensure that the Environmental Monitor and construction personnel inspect Environmentally Sensitive Areas on the Project on a regular basis during any extended work stoppages (e.g., weekends, statutory holidays, Christmas vacation periods) to prevent environmental problems. If potential adverse environmental impacts are observed during work stoppage periods, the Contractor shall initiate and undertake any required environmental measures to avoid or minimize impacts.

165.03 Compliance, Enforcement and Payment

165.03.01 Temporary Pollution Control - The Contractor shall be responsible for implementation of any temporary environmental protection measures, such as pollution control measures. If the Contractor fails to respond, the Ministry may take whatever action is necessary to provide the proper corrective measures. In the event the Contractor fails to take reasonable action to implement temporary environmental control measures, the Ministry Representative or the Environmental Agencies shall determine the need for additional corrective action. The Ministry Representative may deduct incurred costs from any money due or to become due to the Contractor for any corrective action taken in this regard.

165.03.02 Responsibility for Damage to Environment - The Contractor shall bear sole responsibility for any direct or indirect damage to the environment which occurs as a result of failure to comply with these Specifications, the Contract and the directions of the Ministry Representative or recommendations of the Environmental Monitor.

165.03.03 Stop Work Orders - In the event that the Ministry Representative determines that some aspect of the construction is creating or will result in a substantial adverse effect on environmental values or resources on or adjacent to the project site, the Ministry Representative may issue a "stop work order". The Contractor shall subsequently be responsible for advising the Ministry Representative of the intended remedial action. The Ministry Representative may direct the Contractor to cease construction work on an unaffected portion of the site so that any necessary equipment can be immediately diverted to address the environmental emergency. In such an event the Contractor will not be entitled to any claim for compensation from the Ministry.

The Contractor shall be required to modify or halt any aspect of construction or maintenance that the Environmental Agencies determine poses or will pose an environmental concern. The Environmental Agencies may verbally direct the Contractor to modify or halt any such activity or submit formal written instructions to that effect. In either case, the Contractor's immediate compliance with their instructions will be mandatory. The Contractor shall document any such requests by the Environmental Agencies and forward a copy of this documentation to the Ministry Representative within five working days of the initial notification by the Environmental Agencies.

165.03.04 Compliance with Specifications - The Contractor is responsible for ensuring that all subcontractors and employees are in compliance with these Specifications and all applicable environmental legislation and regulatory requirements at all times, and shall take immediate action to rectify problems in this regard. The Contractor shall initiate action to rectify environmental problems. The Contractor shall have available such additional safeguards, safety devices and protective equipment as are necessary to protect the environment. The Contractor shall be responsible for ensuring sufficient safety devices and protective equipment (e.g., pumps, silt fence, armoring, tarps, fuel spill cleanup kits, etc.) are readily available at all times during construction of the Project. The location and inventory of safety devices and protective equipment shall be documented and placed in a prominent location in the Contractor's field office. A copy of this information shall be updated as necessary and provided to the Ministry Representative.

The Environmental Monitor will inspect and monitor conditions at, and in the vicinity of, the project to ensure acceptable levels of pollution and disturbance are not exceeded, and to ensure compliance with the environmental specifications contained herein and in the Contract.

The frequency of such inspection and monitoring efforts shall correspond to the sensitivity and location of construction activities, as well as to the environmental conditions (e.g., increased monitoring and inspection will be conducted during periods of inclement weather). In the event that the Ministry Representative is not satisfied with the frequency and duration of such inspection and monitoring, the Contractor's Environmental Monitor will be required to increase their efforts in this regard to a level satisfactory to the Ministry Representative.

165.03.05 Payment - All the requirements for compliance with these environmental specifications shall be considered incidental to the price bid for the Project under contract and no other compensation shall be made to the Contractor. However, the Contractor may be eligible for reimbursement for the installation of any additional environmental works (e.g., enhancement measures) where directed and approved in writing and not covered in this

Section or elsewhere in the Contract or Schedule of Quantities and Unit Prices. Any eligible reimbursement will be paid for by an Order for Extra Work. The Ministry Manager, in consultation with the Ministry's Environmental Quality staff and the Environmental Agencies, will be responsible for determining the Contractor's eligibility in this regard and for identifying what environmental work is beyond the scope of the Contract and these specifications.

165.04 Erosion, Sediment, and Drainage Control - All works shall be undertaken in a manner that avoids or absolutely minimizes erosion problems and the discharge of siltation or other deleterious substances into any watercourse. The Contractor shall not disturb designated watercourses, unless the express written consent of the Environmental Agencies is obtained prior to initiating construction activities in such areas. No obstruction or debris shall be placed in any watercourse during any operations, unless specified in the Contract or by the Ministry Representative, or for designated watercourses approved by the Environmental Agencies. Should any material be inadvertently placed within the normal high water wetted perimeter of a watercourse, the Ministry Representative shall be notified immediately and the Contractor shall remove the material immediately, using environmentally acceptable construction procedures and under the direction of the Environmental Monitor.

165.04.01 Sediment and Erosion Control - The Contractor shall incorporate all permanent soil erosion control features into the project at the earliest practicable time, as outlined in the accepted work schedule, and shall be responsible for temporary erosion and sediment control measures, including daily inspection of the integrity of such measures during adverse weather conditions or when construction operations are proceeding in Environmentally Sensitive Areas. Control measures shall be capable of continuous operation during working and non-working hours,. Any deficiencies which are observed in erosion control measures shall be immediately corrected.

The Environmental Agencies and the Ministry Representative have the authority to define environmentally sensitive areas or conditions, such as areas of erodible soil, and to direct the Contractor to provide temporary erosion and sediment control measures. These measures may include, but may not be limited to:

- interceptor ditches or berms to direct runoff away from erodible areas;
- slope protection measures such as mulches, hydroseeding, erosion mats, geotextiles, filter fabric, polyethylene covers, or riprap;
- silt fences;
- ditch blocks to reduce flow velocities, and;
- sediment control measures, such as settling ponds.

The Contractor shall have sufficient materials, such as clean rock, granular material, and filter fabric available on-site for emergency protection measures when required.

The Contractor shall regularly maintain sediment and drainage control measures, such that they function as designed. Immediate action shall be taken by the Contractor to correct any deficiency observed in the operation of sediment and erosion control measures. In the event that a deficiency in any sediment or drainage control measures is directly or indirectly creating an adverse environmental impact, the Contractor shall initiate the necessary action to correct the problem within one hour of observing or being informed of the situation. In the event that a deficiency in any sediment or drainage control measures poses the potential to create an adverse environmental impact, the Contractor shall take action within two days of observing or being informed of the situation. In the event that environmental conditions or the status of the situation change in regards to a potential deficiency, the Contractor shall accelerate any proposed correction measure(s) by taking immediate action (i.e., within one hour). The judgment of the need and timing for corrective action in regard to sediment or drainage control measures shall be at the discretion of the Contractor and the Environmental Monitor, and shall be to the satisfaction of the Ministry Representative and the Environmental Agencies.

Sediment and debris accumulations which compromise the functioning of the erosion and sediment control measures shall be removed by the Contractor and disposed of in an environmentally acceptable location and manner. At a minimum, silt fence structures shall be cleaned when sediment accumulation heights exceed 30 cm along any portion of the structure.

In the absence of site-specific requirements from the Fisheries and Oceans Canada or the British Columbia Ministry of Environment, the Contractor shall comply with criteria for suspended solids in runoff water from the site as agreed upon prior to construction by the Environmental Agencies.

Sediment and drainage control methods are detailed in the "Land Development Guidelines" and the MoT publication "Control of Erosion and Shallow Slope Movement". Compliance with the spirit and intent of these guidelines is mandatory.

165.04.02 Sediment and Drainage Management Plans - The Contractor, in consultation with the Contractor's Environmental Monitor, shall be responsible for the preparation and implementation of any Sediment and Drainage Management Plan(s) required for the Project. Such plans will form an essential component of the Contractor's environmental protection program, in terms

of identifying environmentally acceptable construction practices for sediment and erosion controls at specific sites. The Sediment and Drainage Management Plan(s) will be subject to, but not limited to the following requirements:

- Prior to commencement of construction activities on the Project and thereafter when deemed necessary, the Contractor and the Environmental Monitor will identify areas and major construction activities for which the Contractor will be required to prepare one or more Sediment and Drainage Management Plans. Designation of these areas and activities will be done in consultation with the Environmental Agencies and the Ministry's Environmental Quality staff.
- In addition to activities within the Project area, a Sediment and Drainage Management Plan shall be submitted for activities related to the Contractor's work in areas away from the project that have the potential to create adverse environmental impacts, which will be determined by the Contractor in consultation with the Environmental Monitor, the Environmental Agencies and the Ministry Representative. These activities are not confined to the areas immediately adjacent to the right-of-way, but also in other locations in which the work or associated activities will be conducted, including but not limited to haul or access roads, temporary structures, borrow and granular excavations, staging/laydown areas, material storage sites and disposal sites.
- The Contractor shall not commence work in any area so identified until a Sediment and Drainage Management Plan has been submitted and accepted by the Ministry Representative and involved Environmental Agencies.
- The acceptance and approval of the Contractor's Sediment and Drainage Management Plan or updated Plan shall be valid only so long as the conditions and anticipated conditions upon which the Plan and its acceptance were based continue to be applicable.
- The Contractor shall at all times be responsible for ensuring that work is carried out in accordance with a Sediment and Drainage Management Plan or updated Plan, which has been accepted by the Ministry Representative and Environmental Agencies.
- The Ministry Representative or the Environmental Agencies may request the Contractor to update and resubmit its plan(s) at any time, and may require the Contractor to cease work until the update(s) has been submitted and accepted. Whether or not the Ministry Representative or the Environmental Agencies so request, the responsibility for carrying out the work in accordance with a validly accepted Plan lies entirely with the Contractor.
- The Ministry or the Environmental Agencies may order the Contractor to stop work or to take other precautionary or remedial measures whenever the Contractor is carrying out any work that is not in accordance with a Sediment and Drainage Management Plan or updated Plan which has been accepted by the Ministry Representative or the Environmental Agencies, and for which that acceptance is still valid. All costs of any resulting delay shall be borne entirely by the Contractor.

At a minimum, a Sediment and Drainage Management Plan will include the following information and will be prepared in a format satisfactory to the Ministry Representative:

a) A schedule for the proposed activities, as they pertain to the Sediment and Drainage Management Plan and construction staging, including anticipated duration of construction.

b) A description of the construction procedures that will be used to limit the potential for erosion and sediment production, including estimates of work areas where applicable (e.g., volume of material in a proposed stockpile site), description of construction equipment to be used, and staging of operations.

c) A detailed description of the site-specific measures for runoff and drainage management.

d) A detailed description of the site-specific mitigation measures and design information for erosion prevention and control (e.g., sizing of culverts, ditches or sediment/detention ponds, silt fence specifications, and description of ditching and berming).

e) A schedule that identifies the various sloped areas by station and the dates in which they will be protected using temporary and permanent revegetation measures, such as hydroseeding.

f) A detailed description of mitigative measures for ensuring acceptable water quality and quantity at points of discharge to watercourses, including sizes and specifications for any proposed water treatment facilities.

g) A key plan and drawings, in suitable scale and detail, of the site and proposed mitigative measures and applicable construction procedures.

h) A written commitment of on-site equipment (e.g., water pumps) and materials (e.g., silt fence, hay bales, rock armoring, and ditch breakers) for erosion, sediment and drainage control to deal with emergency situations that may arise.

i) A written commitment of on-site equipment and staff to handle any fish salvage and transfer operations, if required.

j) Contact names, positions and telephone numbers.

A description of the specific monitoring procedures prior to, during and after completion of construction activities at this site.

165.04.03 Sediment Control Ponds - The Contractor shall construct sediment control ponds where necessary to prevent the release of unavoidably entrained sediments in runoff from the construction site, and shall construct stormwater detention ponds for the temporary and/or permanent control and discharge of stormwater runoff. The Contractor shall provide designs for and construct sediment control ponds, as well as stormwater detention ponds, on a site-specific basis in a manner that conforms to the "Land Development Guidelines", other than for the following exception(s):

- Sediment control ponds and dry or wet stormwater detention ponds shall be designed and constructed to accommodate the 10-year return period storm under developed conditions with a minimum 0.60 metre freeboard to the top of the berm.

The Contractor is referred to Section 3 of the “Land Development Guidelines” for other pond design criteria and Section 8 of that publication, for sample calculations for water flows and pond sizes.

If during construction the settlement pond is filled to capacity and is still required for sediment control, accumulated settled sediments shall be removed from sediment control ponds and stormwater detention ponds. Such sediments shall be disposed of in an environmentally acceptable location and manner.

165.04.04 Maintenance of Drainage Patterns - Except where interceptor ditches or berms are required to divert sediment laden runoff from the site to a sediment control pond, original drainage patterns shall be maintained throughout construction operations. Interceptor ditches or berms shall be constructed to divert water entering the site away from erosion prone areas. The Contractor will not rely on drainage courses or conduits being shown on the Drawings, and will make whatever investigation is necessary. Where a natural watercourse traversing the construction site crosses this interceptor ditch or berm, an armoured or lined ditch or a culvert shall be installed in order to pass accumulated flows through or around the construction site in a manner that maintains the natural runoff pattern, unless otherwise directed by the Environmental Agencies.

All ditches constructed for interception of clean water outside the work site(s) to divert it around the work site(s) and for collection of treated water from the work site(s) shall be designed and constructed to accommodate the 10-year return period storm with a 0.3 m freeboard under developed conditions. Temporary and/or permanent drainage ditches, including those constructed for sediment and erosion control shall be designed to prevent high water velocities and erosion by including measures such as check dams, drop structures or erosion-resistant liners or armoring.

165.04.05 Storm Drainage Systems - In order to minimize siltation, the Contractor may be required to block storm drain inlets, or to activate inlets by means of sandbags, berms or swales, as circumstances require. Berms shall be constructed of clean, non-erodible granular material. The Contractor shall maintain, on a regular basis, any inlets activated during the course of construction. Any deficiencies noted in these works shall be corrected immediately in the event that the inlet is plugged and within the same day if the site is relatively dry.

The Contractor shall ensure adequate inspection and correction of any storm drainage system deficiencies for sites within the influence of the Project. During periods of inclement weather, the Contractor shall ensure project staff inspect storm drainage systems throughout the day, and if necessary, make arrangements for inspection and maintenance during regular work stoppage periods, such as evenings or weekends. At a minimum, such inspections shall be carried out at the start of the regular workday, at mid-day in the workday, and one hour prior to end of the regular workday. Furthermore, in the event of flooding problems, the Contractor shall be responsible for any damages or compensation resulting from impacts on residences or businesses caused by a failure or deficiency in the Contractor’s design and maintenance of the storm drainage system.

During concrete curb and gutter construction, road surface runoff shall be directed by berms or swales away from concrete which has been poured within the previous 48 hours.

165.05 Clearing and Grubbing

165.05.01 Limits for Clearing and Grubbing - Clearing and/or grubbing shall extend only to the designated limits, as defined in the Contract and the Design and marked on site The Contractor may be required to use close-cut, no grub practices (i.e., cutting trees at ground level and not removing root system) or undertake hand clearing at environmentally sensitive sites, as designated in the Contract or in the Design.

The Contractor shall prepare a detailed clearing and

grubbing plan in consultation with the Environmental Monitor. The Contractor will obtain approval for this plan from the Environmental Agencies and the Ministry Representative and any necessary tree removal permits from the Ministry of Forests and/or municipal agencies prior to initiating any clearing and grubbing operations within 50 m of any designated watercourse or Environmentally Sensitive Area.

The Contractor shall minimize ground disturbance in the vicinity of any watercourse or Environmentally Sensitive Area, and shall not proceed with clearing and grubbing closer than 50 m from a designated watercourse or Environmentally Sensitive Area prior to marking the limits of the Fisheries Sensitive Zone or Environmentally Sensitive Area. The width of these zones, if not specified in the Contract, will be determined by the Designer in consultation with the Environmental Agencies. The Contractor shall mark the limits of any Fisheries Sensitive Zone or Environmentally Sensitive Area, such as orange flagging tape or orange snow fence, at a minimum height of 1.5 m.

All clearing and grubbing activities in this zone must be granted prior approval by the Environmental Agencies. The Contractor shall not conduct approved grubbing operations within this zone until ready to proceed with earthwork and stabilization.

In areas near designated watercourses or other Environmentally Sensitive Areas determined by the Environmental Agencies to be of high erosion or siltation potential, the surface area of erodible soil exposed at one time may be limited. This will supersede DBSS 200.03 and DBSS 200.04. Where exposed materials are sensitive to erosion, the Contractor shall confine grubbing to an area where excavation and earthwork is to be actively conducted within approximately 30 days following the completion of grubbing operations. In the event that a highway project consists only of clearing and grubbing operations, the cleared ground shall be protected and able to withstand specified periods of inclement weather.

165.05.02 Protection of Vegetation - The Contractor shall protect all vegetation growing outside of the designated and marked areas for clearing and grubbing, as specified in DBSS 769 "Protection and Retention of Vegetation", with the exception of danger trees, which must be hand-felled and removed with minimum disturbance to retained vegetation.

165.05.03 Clearing Activities - Falling of timber into a watercourse is prohibited except in situations in which safety considerations dictate otherwise or unless approved by the Environmental Monitor. Any trees or large pieces of woody debris that accidentally fall into a watercourse and require removal shall be removed in a manner that minimizes the disturbance of the watercourse and adjacent banks,. Skidding of logs across watercourses will not be permitted. The Environmental Monitor must be present during removal of any large woody debris or trees from watercourses.

165.05.04 Disposal of Combustible Materials - The Contractor shall comply with the Forest Act and the Open Burning Smoke Control Regulation under the British Columbia Waste Management Act. For additional information regarding the regulation, the Contractor should contact regional Environmental Protection staff in the British Columbia Ministry of Environment office and the Fire Control staff in the British Columbia Ministry of Forests.

The Contractor shall prevent heat or smoke damage to all vegetation that has been designated for preservation. The use of waste oil and/or tires as fire accelerators shall not be permitted. The Contractor may be required to use a forced air method of burning. Burning piles shall not be located within 50 m of any watercourse, wetland or other Environmentally Sensitive Area.

In situations in which the material generated as a result of clearing and/or grubbing may not be burned on site, the Contractor shall dispose of it in a manner approved by the Environmental Agencies who may require approval of the British Columbia Ministry of Environment.

165.06 Stripping Operations

165.06.01 Exposure of Erodible Earth - In areas where erosion or siltation is anticipated, the duration of exposure of erodible earth material shall be minimized. In such cases the surface area of erodible earth material exposed at one time shall be determined by the Environmental Monitor, to the satisfaction of the Environmental Agencies and the Ministry Representative. (See DBSS 165.05.01 Limits for Clearing and Grubbing)

165.06.02 Placement of Stripped Material - The Contractor shall avoid placement of stripped materials in areas adjacent to watercourses or other Environmentally Sensitive Areas. Sediment and erosion control measures must be taken prior to and after placement of stockpiles of stripped material in areas where natural drainage or storm water could erode the stockpile and thereby transport pollutants to surface waters. The location of stockpiles and erosion control measures must be approved by the Environmental Monitor, to the satisfaction of the Environmental Agencies and the Ministry Representative prior to placement of stripped materials and may be specified in the Contract. The Contractor shall ensure that all stockpiles are stable.

Polyethylene sheeting or other suitable tarp material shall

be used to cover temporarily exposed steep surfaces or stockpiles of erodible materials, such as topsoil, sand, gravel or roadbase fill. The Contractor shall ensure such erodible materials are properly covered with sheeting or suitable tarp materials immediately after creation of any temporary stockpiles. Such sheeting or tarp materials shall be examined and maintained on a regular basis. The sheeting or tarp materials shall be sufficiently anchored to prevent displacement by winds.

165.06.03 Slope Protection - The Contractor shall employ suitable techniques to prevent the initiation of surface soil erosion and movement of sediments from slopes, particularly those exposed as a result of clearing, grubbing and stripping operations. Acceptable techniques and conditions of implementation of erosion and sediment control measures for slopes are outlined in Section 3 (i.e., entitled “Slope Protection and Surface Protection”) of the “Land Development Guidelines” and the MoT publication "Manual of Control of Erosion and Shallow Slope Movement". Compliance with the requirements specified in this Section, as well as any Authority/MoT manual(s) developed specifically for the Project, will be mandatory.

165.07 Earthwork Operations (Subgrade Construction)

165.07.01 Excavation and Disposal of Waste or Surplus Material - The creation and use of any site on Crown or private lands for the placement and disposal of waste or surplus material requires prior approval from the Ministry Representative and may require the approval of representatives of British Columbia Forests and the Environmental Agencies. Unless another time period is specified in the Contract or by the Ministry Representative, not less than fifteen days prior to disposing of any waste or surplus material, the Contractor shall submit to the Ministry Representative for approval, a proposal delineating the locations and extent of the areas in which the Contractor intends to dispose of such material. The proposal shall describe the nature of the material and the methods to be employed in material placement, stabilizing and site revegetation. If a disposal site is required off the right-of-way, the proposal shall also indicate the procedures for any required land clearing activity. No waste or surplus material shall be disposed of until the proposal has been approved. All work shall be done in accordance with the approved proposal.

All waste disposal sites shall be selected such that spoil is prevented from entering any watercourse. The excavated material shall be properly drained, spread and trimmed to a stable slope not exceeding 1.5 to 1, in a manner which minimizes disturbance of watercourses and vegetated areas. All waste sites shall be revegetated immediately after creation of the waste disposal site, or else suitable temporary erosion control measures, such as tarps, shall be used until revegetation is undertaken. The Contractor is responsible for ensuring that revegetation is successfully accomplished unless otherwise specified in the Contract.

165.07.02 Work in Areas of High Water Table - Work in areas of high water table shall be scheduled to proceed during dry weather periods unless otherwise specified in the Contract. The Contractor shall adopt all necessary mitigation measures to avoid or minimize adverse impacts in such areas.

165.07.03 Drilling and Blasting - Blasting operations shall be conducted in a manner that minimizes the disturbance of residences, businesses and public infrastructure (e.g., schools and churches) and to aquatic and terrestrial habitats and organisms due to shock waves, noise and vibration.

The Contractor shall not conduct blasting within 400 m of fish habitat without the approval of the blasting plan by the Fisheries and Oceans Canada; British Columbia Ministry of Environment Fish and Wildlife officers. The Environmental Agencies may require the Contractor to undertake mitigative measures such as scheduling of works in Timing Windows given by the Environmental Agencies, smaller staggered blasts or use of blasting mats to minimize impacts. Vibrations and shock waves in fisheries habitat resulting from blasts shall comply with the Fisheries and Oceans Canada "Guidelines for the Use of Explosives in or Near Canadian Fisheries Waters".

Blasting operations shall be controlled to minimize the entry of fly rock into watercourses or adjacent to the right-of-way, particularly in an urban setting. If required by the Environmental Agencies, the Contractor shall comply with the Agencies' directions in removal of fly rock from watercourses and Environmentally Sensitive Areas at no cost to the Ministry.

When blasting is to be conducted adjacent to or in a designated environmentally sensitive area, the Contractor shall provide at least 24 hours notice of each blast to the Ministry Representative and to the Environmental Agencies. The Contractor shall provide at least 24 hours written notice to those individuals or groups potentially affected by shock waves, noise and vibration from blasting operations.

165.08 Borrow and Sand / Gravel Pits - All Borrow and Sand/Gravel pit locations, operations, reclamation and revegetation shall comply with the Ministry of Energy and Mines / MoT / Natural Resources Canada joint publication "Reclamation and Environmental Protection Guidelines for Sand, Gravel and Quarry Operations in British Columbia".

165.08.01 Control of Drainage - Prior to borrow excavations, or quarry, sand or gravel pit development, all

surface water entering the site shall be controlled so that its entrance into the operating area is prevented. Drainage shall be controlled throughout borrow operations in order to prevent natural drainage and surface runoff from carrying sediment into adjacent watercourses. The Contractor shall prepare a Sediment and Drainage Management Plan (see DBSS 165.04.02) prior to the creation and/or operation of any borrow excavations or quarry, sand or gravel pit development sites. The Contractor shall also prepare a Reclamation Plan.

165.08.02 Location and Development of Borrow and Sand/Gravel Pits - Clearing for borrow and sand/gravel pits shall comply with DBSS 165.05 - Clearing and Grubbing. All topsoil shall be stockpiled for future reclamation.

Borrow excavations shall not be located in the wetted perimeter of a watercourse or on a stream floodplain at a location likely to create a new channel to the stream at a time of flooding, without the approval of the Environmental Agencies. Borrow sources located within 100 metres of the wetted perimeter of a watercourse or at distances less than 100 m from any water-impounding dyke or dam embankment will require the design of a Professional Engineer or Professional Geologist. Final floor elevations shall be sufficiently high that they will not be subject to fluctuations in the groundwater table.

In erosion-prone areas, overburden removal shall be restricted to an area that will be excavated within one year. The open face(s) of the borrow pit should be no larger than necessary for efficient pit operation. Material should be exhausted at one location in a pit prior to the opening of a new face to remove a similar type and grade of material.

The Contractor shall be responsible for maintaining the functionality of erosion, sediment and drainage control measures within and around borrow and sand/gravel pits throughout all phases of overburden and material removal, as well as during periods of work stoppages. Borrow and/or sand/gravel pits shall be rehabilitated and decommissioned in an environmentally acceptable manner and to the satisfaction of the Ministry Representative and the Environmental Agencies.

To minimize the spread of noxious weed seeds between construction sites and gravel pits/quarries, all crushing equipment and mining vehicles must be cleaned with a power/high pressure washer (manually cleaned in freezing temperatures), before entering any MoT pit or quarry. If the pit contains noxious weeds the crushing equipment and mining vehicles must also be cleaned with a power/high pressure washer (manually cleaned in freezing temperatures) before leaving the pit/quarry. Prior to the production of aggregates the pit floor, crusher/plant location, equipment-parking area, stockpile site(s), pit face(s) and area to be mined will be inspected by the contractor for the presence of noxious weeds. If noxious weeds are present they must be removed mechanically by the contractor. Noxious weeds are defined as non-native plants species designated on the provincial or regional district lists as defined by the Weed Control Act.

165.08.03 Washing Operations - In locations where siltation of a fish-bearing watercourse may occur, approval shall be obtained from the Environmental Agencies prior to the establishment of aggregate washing operations, and all wash water shall be subject to DBSS 165.04.

Waste or surplus material shall be disposed of at a waste disposal site approved by the Ministry and the Environmental Agencies. The Contractor shall create and designate, with appropriate signing, suitable sites and facilities for disposal of waste or surplus material. In particular, the disposal of deleterious materials, such as surplus concrete or concrete wash water in sites other than those approved by the Environmental Monitor and the Environmental Agencies will be prohibited. The Contractor and the Contractor's Environmental Monitor shall be responsible for ensuring all construction personnel, including subcontractors, are made aware of this requirement. In the event that it is not feasible to install a suitable concrete waste disposal site immediately adjacent to the work site, such as during a concrete pour on a long bridge deck, then the Contractor shall take the necessary precautions to isolate the work site from any watercourse or other environmentally sensitive area and to temporarily collect the concrete waste or wash water and transport it to a nearby suitable disposal facility.

165.09 Detour, Access and Haul Roads

165.09.01 Design, Construction and Operation of Access Roads - Any temporary access, detour and/or haul roads associated with the project shall be constructed to accommodate all required uses and maintained throughout the course of construction operations in a safe, environmentally sound manner.

The location, alignment, design and construction of all detour, access and haul roads shall be subject to the approval of the Ministry and the appropriate Environmental Agencies. Clearing shall extend no further than safe operation requires, or to the edge of cut or toe of fill. Root systems of cleared vegetation shall not be removed except in the area of cuts or actual road surface.

The Contractor shall employ suitable measures to maintain air quality, visibility, and safe conditions in the use of access, detour and/or haul roads associated with the Project.

165.09.02 Drainage and Erosion Control - Drainage